



3 Seismology and Earth Structure

Ordinary language undergoes modification to a high pressure form when applied to the interior of the earth; a few examples of equivalents follow:

Ordinary meaning:

*dubious
perhaps
vague suggestion
trivial objection
uncertain mixture of all the elements*

High pressure form:

*certain
undoubtedly
positive proof
unanswerable argument
pure iron*

Francis Birch, 1952

3.1 Introduction

A major application of seismology is the determination of the distribution of seismic velocities, and hence elastic properties, within the earth. This distribution, known as *earth structure*, gives the basic constraint on the mineralogical, chemical, and thermal state of the earth's interior. Seismological data are important for this purpose because their resolving power is generally superior to that of other geophysical methods. For example, although gravity and magnetic data indicate the presence of a dense fluid core at depth, they provide only relatively weak constraints on its density and size. By contrast, seismological data indicate the depth of the core-mantle boundary and the sharp change in properties that occurs there. Above the boundary, both *P* and *S* waves propagate in the solid mantle, whereas in the liquid outer core no *S* waves propagate and the *P*-wave velocity drops sharply. The observed velocities are the primary basis for our models of the physical properties and chemical composition of the material on either side of this boundary. Similarly, the distinction between the crust and the mantle and many inferences about their structure and composition come from seismological observations. More generally, by establishing the essentially layered structure of the earth, seismology provides the primary evidence for the process of differentiation whereby material within planets became compositionally segregated during their evolution. As a result, many crucial issues about the other terrestrial planets could be resolved if seismological data were available.

Constraints from seismology are crucial for other disciplines of the earth sciences, and vice versa. Seismology gives *earth models* describing the distribution of *P*- and *S*-wave velocities and density. Going from an earth model to a description of the chemical, mineralogical, thermal, and rheological state of the earth's interior requires additional information. There are thus two types of uncertainty in our knowledge of the earth's interior. In some cases, such as the structure of the inner core, the seismological results are still under discussion. In others — for example, the nature of the 660 km discontinuity in the mantle — the basic seismological results are generally accepted, but their mineralogic and petrologic interpretations remain under investigation. Given our scope here, we only summarize the implications of seismological data for models of the earth's interior.

The fundamental data for seismological studies of the earth's interior are the travel times of seismic waves. The measurements available are the arrival times of seismic waves at receivers. To convert these to travel times, the origin time and location of the source must be known. These parameters, which are known for artificial sources, must be estimated from the observations for earthquake sources. Hence travel time data include information about both the source and the properties of the medium, and separating the two is a challenge in many seismological studies.

The travel times are used to learn about the velocity structure between the source and the receiver. As we saw in the last chapter, waves follow paths that depend on the velocity

structure. Hence the structure must be known to find the paths that the waves took. To illustrate this, consider the travel time between two points. If the velocity were constant, the ray path would be a straight line, and the velocity could be found by dividing the distance by the travel time. If, instead, an interface separates media with different velocities, the ray path would consist of two line segments, depending on the velocities, and the travel time would be the sum of the time spent along each segment. For a more complicated velocity distribution, the ray path would also be more complicated.

This problem can be posed mathematically by writing the travel time between the source (s) and receiver (r) as the integral of $1/\text{velocity}$, or slowness, along the ray path

$$T(s, r) = \int_s^r \frac{1}{v(x)} dx. \quad (1)$$

In simple cases, where the ray path is a set of segments with constant velocity, the integral is just a sum over the time in each segment. Thus the travel time gives an integral constraint on the velocity distribution between the source and the receiver, but does not indicate which of the many paths satisfying the constraint the ray followed. As a result, an individual measurement is inadequate to show the distribution of velocities. Fortunately, as we shall see, a set of travel times between different sources and receivers provides much more information. In addition, useful information is derived from the amplitudes and waveforms of seismic waves.

This example illustrates an interesting feature of determining velocity structure from travel times. If the velocity structure is known, the forward problem of finding the travel times and amplitudes is straightforward. However, the inverse problem of using the travel times and amplitudes measured at the surface to find the velocity structure at depth is more difficult, and various methods are used. For example, in addition to using travel times directly, we have seen that velocity structure is studied using the dispersion of surface waves (Section 2.8) and the eigenfrequencies of normal modes (Section 2.9), quantities that correspond to travel times.

In this chapter, we follow the approach discussed in Section 1.1.2 of treating the earth with a series of progressively more complex and, hopefully, more accurate models. We begin with the homogeneous, isotropic, elastic, layered halfspace used in Chapter 2 to derive seismic wave propagation. This approximation of uniform flat layers is often used in crust and upper mantle studies, where the distance between source and receiver is less than a few hundred kilometers. We then consider larger source–receiver distances, for which spherical geometry is required, and then the anisotropic and anelastic behavior of the earth. Throughout these discussions, we will see that although velocity varies primarily with depth, there are important lateral variations, or heterogeneities. Finally, we consider the implications of the observed heterogeneous, anisotropic, and anelastic

velocity structure for the composition of the earth. Later, in Chapter 7, we discuss further how seismic data can be used to study laterally variable velocity structure.

3.2 Refraction seismology

3.2.1 Flat layer method

The simplest approach to the inverse problem of determining velocity at depth from travel times treats the earth as flat layers of uniform-velocity material. We thus begin by deriving the travel time curves for such a model, which show when seismic waves arrive at a particular distance from a seismic source. The travel times, especially those of waves that are critically refracted at the interfaces, are used to find the velocities of the layers and underlying halfspace and the layer thicknesses. As a result, this technique is called *refraction seismology*.

Refraction seismology is used on vastly differing scales. Near-surface structure at depths less than 100 meters can be studied using a sledge hammer or a shotgun as a source and a single receiver. Similar methods are used to study the crust and the upper mantle, with earthquake or explosion sources and many receivers at distances of hundreds of kilometers.

The simplest situation, shown in Fig. 3.2-1, is a layer of thickness h_0 , with velocity v_0 , overlying a halfspace with a higher velocity, v_1 . We write the velocities as “ v ” to indicate that the analysis applies for either P or S waves. There are three basic ray paths from a source on the surface at the origin to a surface receiver at x . The travel times for these paths can be found using Snell’s law.

The first ray path corresponds to a *direct wave* that travels through the layer with travel time

$$T_D(x) = x/v_0. \quad (1)$$

This travel time curve (Fig. 3.2-2) is a linear function of distance, with slope $1/v_0$, that goes through the origin.

The second ray path is for a wave reflected from the interface. Because the angles of incidence and reflection are equal,

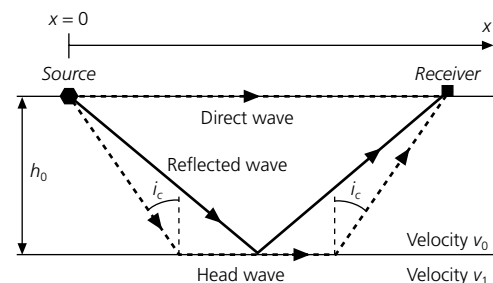


Fig. 3.2-1 Three basic ray paths for a layer over a halfspace model. The direct and reflected rays travel within the layer, whereas the head wave path also includes a segment just below the interface. For the head wave to exist, the layer velocity v_0 must be less than the halfspace velocity v_1 .